

PETER FUNG

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EDUCATION

National University of Singapore

Aug 2021 – Jun 2025

Bachelor of Computing (Honors) in Computer Science (GPA: 4.16/5.0)

Specializations in Software Engineering and Cybersecurity

WORK EXPERIENCE

NCS, AI Software Engineer

Sep 2025 – Present

- Built FastAPI backend services and LLM pipelines for NCSgpt, an internal AI platform used by 1,000+ employees.
- Developed a Retrieval-Augmented Generation (RAG) pipeline over enterprise knowledge sources, combining vector search with LLM generation to ground NCSgpt responses and reduce hallucinations.
- Instrumented NCSgpt with Langfuse and OpenTelemetry for end-to-end LLM observability across prompt tracing, latency monitoring, and cost tracking.
- Designed an Email Agent harness over Microsoft Graph APIs with intent routing and retrieval, supporting secure email search, summarisation, and reply drafting.
- Engineered an AI PowerPoint generation service running Dockerised Python agents that autonomously generate, modify, and adapt decks.

Ryde, Frontend Engineer Intern

Jan 2025 – Jun 2025

- Built Ryde's WeChat Mini Program frontend from scratch as the sole frontend developer.
- Developed real-time one-on-one chat with WebSockets, delivery statuses, read receipts, and push notifications.
- Integrated in-app calling via WebView, handling call initiation, join, and termination states.
- Reduced initial chat history load time by ~60% using lazy-loading pagination.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#, Dart, HTML, CSS

LLM & AI: OpenAI SDK, Anthropic SDK, Langfuse, RAG, Prompt Engineering, LLM Tool Calling

Backend & APIs: FastAPI, RESTful APIs, WebSocket APIs, Microservices

Frontend & Mobile: React, Flutter, Tailwind CSS, WeChat Mini Programs

Cloud & Infrastructure: Microsoft Azure, Azure Kubernetes Service (AKS), Docker, Terraform

Databases: PostgreSQL, MySQL

PROJECTS

PayMeLah, Group Bill Splitting Application

<https://paymelah-app.vercel.app/> | <https://apps.apple.com/sq/app/paymelah/id6761058776>

- Built a React Native and FastAPI bill-splitting app for receipt scanning and shared expense tracking.
- Developed an OCR and AI receipt parsing pipeline, reducing manual input time by over 70%.
- Implemented itemised cost assignment, shared-item splitting, and automated settlement calculations.

HomeCooked, AI-Powered Recipe Creation Application

<https://home-cooked-gamma.vercel.app/>

- Built a React and Next.js app for recipe management, scraping, and AI-assisted recipe creation.
- Used OpenAI APIs to convert web and Instagram content into structured recipes in under 30 seconds.

